1 $3(3x-1)=4(x+5)$ or $9x-3=4x+20$ **[M1]**

$5x=23$ **[M1]**

4.6 or $\frac{23}{5}$ **[A1]**

[Total Marks: 3]

2 Multiplies all terms by 12 (or forms a single fraction): $3(2x-3)+4(x+1)=24$ **[M1]**

Expands brackets correctly: $6x-9+4x+4=24$ **[M1]**

2.9 or $\frac{29}{10}$ **[A1]**

[Total Marks: 3]

3 Valid method to eliminate one variable, e.g. multiplying the first equation by 2 to get $4x-2y=14$ and adding the equations **[M1]**

$x=4$ **[A1]**

$y=1$ **[A1]**

[Total Marks: 3]

4 Multiplies one or both equations to equate coefficients of x or y (e.g. $4x-10y=44$) **[M1]**

Adds or subtracts equations to eliminate one variable (e.g. $13y=-39$) **[M1]**

$x=3.5$, $y=-3$ **[A1]**

[Total Marks: 3]

5 (a) $4x-12=2x+10$ **[M1]**

$x=11$ **[A1]**

(b) $5y=2x-1$ **[M1]**

$\frac{5y+1}{2}$ **[A1]**

[Total Marks: 4]

6 (a) $6x-3=5x+8$ **[M1]**

11 **[A1]**

(b) $y+7=4x$ or $\frac{y}{4}=x-\frac{7}{4}$ **[M1]**

$\frac{y+7}{4}$ or $\frac{y}{4}+\frac{7}{4}$ **[A1]**

[Total Marks: 4]

7 **(a)** M1 for expanding brackets correctly: $6x - 15 = 4x + 4$

M1 for correctly isolating x terms, e.g., $6x - 4x = 4 + 15$ or $2x = 19$

9.5 or $\frac{19}{2}$ **[A1]**

(b) M1 for multiplying by 3 to get $3V = \pi r^2 h$

$\frac{3V}{\pi r^2}$ **[A1]**

[Total Marks: 5]

8 **(a)** $6x - 3 = 5x + 8$ **[M1]**

11 **[A1]**

(b) $5y = 2x + 3$ **[M1]**

$2x = 5y - 3$ **[M1]**

$\frac{5y - 3}{2}$ **[A1]**

[Total Marks: 5]

9 **(a)** $8x - 12 = 5x + 9$ **[M1]**

$3x = 21 \implies x = 7$ **[A1]**

(b) Correctly eliminates one variable: e.g. $9x + 6y = 48$ and $10x - 6y = 28$ **[M1]**

$19x = 76 \implies x = 4$ **[A1]**

$y = 2$ **[A1]**

[Total Marks: 5]

10 **(a)** Expands brackets correctly to $6x - 3 = 4x + 11$ **[M1]**

7 **[A1]**

(b) Multiplies second equation by 2 to get $2x - 2y = 4$ OR substitutes $x = y + 2$ into first equation **[M1]**

Correctly finds $x = 4$ **[A1]**

Correctly finds $y = 2$ **[A1]**

[Total Marks: 5]

11 **(a)** $5x - 3 = 2x + 12$ **[M1]**

5 **[A1]**

(b) Multiply by 12 to get $4(x + 2) + 3(2x - 1) = 36$ (or equivalent common denominator method) **[M1]**

$4x + 8 + 6x - 3 = 36$ **[M1]**

$\frac{31}{10}$ or 3.1 **[A1]**

[Total Marks: 5]

12 (a) Expands bracket to obtain $6x - 15$ **[M1]**

Rearranges terms to isolate x, e.g., $6x - 4x = 7 + 15$ or $2x = 22$ **[M1]**

11 **[A1]**

(b) Multiplies by 5 to obtain $5y = 2x + 3$ **[M1]**

$\frac{5y - 3}{2}$ **[A1]**

[Total Marks: 5]

13 (a) $8x - 4 = 3x + 11$ **[M1]**

$5x = 15$ **[M1]**

3 **[A1]**

(b) $2y = 3x - 5$ **[M1]**

$2y + 5 = 3x$ **[M1]**

$\frac{2y + 5}{3}$ or $\frac{2y}{3} + \frac{5}{3}$ **[A1]**

[Total Marks: 6]

14 (a) $5x - 10 = 3x + 14$ **[M1]**

$2x = 24$ **[M1]**

12 **[A1]**

(b) $2A = h(a + b)$ or $\frac{A}{h} = \frac{a + b}{2}$ **[M1]**

$\frac{2A}{h} = a + b$ or $2A = ha + hb$ **[M1]**

$\frac{2A}{h} - a$ or $\frac{2A - ah}{h}$ **[A1]**

[Total Marks: 6]

15 (a) $6x - 15 = 4x + 7$ **[M1]**

$2x = 22$ **[M1]**

11 **[A1]**

(b) Equates coefficients, e.g. multiplies first equation by 2 and second equation by 3 to get $4x + 6y = 26$ and $15x - 6y = 12$ **[M1]**

Adds equations to eliminate y, e.g. $19x = 38$, so $x = 2$ **[A1]**

$y = 3$ **[A1]**

[Total Marks: 6]

16 **(a)** Expands brackets: $4x - 12 = 2x + 7$ **[M1]**

Rearranges to collect terms: $2x = 19$ **[M1]**

9.5 or $\frac{19}{2}$ **[A1]**

(b) Multiplies first equation by 2 to equate coefficients of y: $10x - 4y = 32$ (or equivalent valid method) **[M1]**

Adds equations to eliminate y: $13x = 39 \implies x = 3$ **[A1]**

Substitutes $x = 3$ into either equation to find y: $15 - 2y = 16 \implies y = -0.5$ **[A1]**

[Total Marks: 6]

17 $y(x + 4) = 3x - 2$ **[M1]**

$xy - 3x = -4y - 2$ or $3x - xy = 4y + 2$ **[M1]**

$x(y - 3) = -4y - 2$ or $x(3 - y) = 4y + 2$ **[M1]**

$\frac{4y + 2}{3 - y}$ or $\frac{-4y - 2}{y - 3}$ **[A1]**

[Total Marks: 4]

18 $y(x - 5) = 2x + 3$ or $xy - 5y = 2x + 3$ **[M1]**

$xy - 2x = 5y + 3$ or $2x - xy = -5y - 3$ **[M1]**

$x(y - 2) = 5y + 3$ or $x(2 - y) = -5y - 3$ **[M1]**

$\frac{5y + 3}{y - 2}$ or $\frac{-5y - 3}{2 - y}$ **[A1]**

[Total Marks: 4]

19 Multiplies by the denominator: $y(2x + 1) = 3x - 5$ **[M1]**

Expands the bracket: $2xy + y = 3x - 5$ **[M1]**

Rearranges to collect all terms involving x on one side: $y + 5 = 3x - 2xy$ (or $2xy - 3x = -y - 5$) **[M1]**

$\frac{y + 5}{3 - 2y}$ or $\frac{-y - 5}{2y - 3}$ **[A1]**

[Total Marks: 4]

20 **(a)** Multiplies equations to equate coefficients of x or y, e.g., $9x + 6y = 36$ and $10x - 6y = 2$ **[M1]**

Adds or subtracts equations to eliminate a variable **[M1]**

$x = 2$ and $y = 3$ **[A1]**

(b) $(2x \pm a)(x \pm b) = 0$ where $ab = 12$ **[M1]**

$(2x + 3)(x - 4) = 0$ **[M1]**

$x = 4$ and $x = -1.5$ (or $-\frac{3}{2}$) **[A1]**

[Total Marks: 6]

21 (a) Correct method to equate coefficients (e.g., $10x + 15y = 65$ and $10x - 4y = 8$, leading to $19y = 57$) or correctly substitute **[M1]**

$y = 3$ **[A1]**

$x = 2$ **[A1]**

(b) $y(x + 2) = 3x - 1$ **[M1]**

$xy + 2y = 3x - 1$ **[M1]**

$xy - 3x = -1 - 2y$ or $2y + 1 = 3x - xy$ **[M1]**

$\frac{-1 - 2y}{y - 3}$ or $\frac{2y + 1}{3 - y}$ **[A1]**

[Total Marks: 7]

22 (a) $y(x + 4) = 3x - 2$ **[M1]**

$xy + 4y = 3x - 2$ **[M1]**

$xy - 3x = -4y - 2$ or $x(y - 3) = -(4y + 2)$ **[M1]**

$\frac{-4y - 2}{y - 3}$ or $\frac{4y + 2}{3 - y}$ **[A1]**

(b) $5(x + 1) = 2(x - 2)$ **[M1]**

$5x + 5 = 2x - 4$ **[M1]**

-3 **[A1]**

[Total Marks: 7]

23 (a) Multiplies equations to equate coefficients, e.g., $9x + 6y = 36$ and $10x - 6y = 2$ **[M1]**

Adds or subtracts equations to eliminate one variable, e.g., $19x = 38$ **[M1]**

$x = 2$ **[A1]**

$y = 3$ **[A1]**

(b) Finds correct product and sum components, e.g., $-10x$ and $3x$ **[M1]**

Correctly factorises to $(2x + 3)(x - 5) = 0$ **[M1]**

$x = -1.5, x = 5$ **[A1]**

[Total Marks: 7]

24 (a) Multiply the second equation by 2 to get $8x + 2y = 36$ (or correct substitution method)

[M1]

$x = 5$ **[A1]**

$y = -2$ **[A1]**

(b) $T(p-2) = 3p+4$ **[M1]**

$Tp - 2T = 3p + 4$ **[M1]**

$Tp - 3p = 2T + 4$ or $p(T-3) = 2T+4$ **[M1]**

$\frac{2T+4}{T-3}$ or $\frac{-2T-4}{3-T}$ **[A1]**

[Total Marks: 7]

25 **(a)** Multiplies the first equation by 2 to give $4x - 2y = 18$ (or rearranges to $y = 2x - 9$) **[M1]**

Adds the equations to eliminate y, yielding $7x = 28$ (or correctly substitutes to form an equation in one variable) **[M1]**

$x = 4$ **[A1]**

$y = -1$ **[A1]**

(b) Multiplies throughout by a common denominator (e.g., 12) to obtain $4(2x) - 3(x-4) = 30$ **[M1]**

Expands brackets correctly: $8x - 3x + 12 = 30$ **[M1]**

$5x = 18$ **[M1]**

3.6 or $\frac{18}{5}$ **[A1]**

[Total Marks: 8]

26 **(a)** Multiply second equation by 2: $4x - 10y = 18$ **[M1]**

Subtract equations to eliminate x: $13y = -13$ **[M1]**

$y = -1$ **[A1]**

$x = 2$ **[A1]**

(b) $(3x+2)(x-4) = 0$ **[M2]**

If M0, M1 for $(3x \pm a)(x \pm b)$ where $ab = 8$ or $a \pm 3b = -10$

$x = -\frac{2}{3}$ **[A1]**

$x = 4$ **[A1]**

[Total Marks: 8]

27 Forms common denominator or multiplies through: $2(x-2) + 1(x-1) = (x-1)(x-2)$ **[M1]**

Expands brackets correctly: $2x - 4 + x - 1 = x^2 - 3x + 2$ **[M1]**

Rearranges to a 3-term quadratic: $x^2 - 6x + 7 = 0$ **[A1]**

Applies quadratic formula: $\frac{-(-6) \pm \sqrt{(-6)^2 - 4(1)(7)}}{2(1)}$ or completes the square: $(x-3)^2 - 9 + 7 = 0$ **[M1]**

Correct unsimplified surd form: $\frac{6 \pm \sqrt{8}}{2}$ or $(x-3)^2 = 2$ **[M1]**

$3 \pm \sqrt{2}$ **[A1]**

[Total Marks: 6]

28 (a) Multiplies through by $(x-1)(2x+1)$: $4(2x+1) - 3(x-1) = 2(x-1)(2x+1)$ **[M1]**

Expands left-hand side: $8x + 4 - 3x + 3 = 5x + 7$ **[B1]**

Expands right-hand side: $2(2x^2 - x - 1) = 4x^2 - 2x - 2$ **[B1]**

Rearranges cleanly to $4x^2 - 7x - 9 = 0$ with no errors **[A1]**

(b) Substitutes into quadratic formula: $x = \frac{-(-7) \pm \sqrt{(-7)^2 - 4(4)(-9)}}{2(4)}$ **[M1]**

Evaluates discriminant: $49 + 144 = 193$ **[A1]**

$\frac{7 + \sqrt{193}}{8}$ and $\frac{7 - \sqrt{193}}{8}$ (or $\frac{7 \pm \sqrt{193}}{8}$) **[A1]**

[Total Marks: 7]

29 (a) Squares both sides and clears denominator: $y^2(t+2) = 3t - 5$ **[M1]**

Collects t terms on one side: $2y^2 + 5 = 3t - ty^2$ or $t(y^2 - 3) = -5 - 2y^2$ **[M1]**

$t = \frac{2y^2 + 5}{3 - y^2}$ or $t = \frac{-5 - 2y^2}{y^2 - 3}$ **[A1]**

(b) One correct root seen: -1.69, -1.00, or 2.69 **[B1]**

All three roots correct to 3 significant figures: -1.69, -1.00 (or -1), 2.69 **[B2]**

[Total Marks: 6]

30 (a) B1 for $x + y = 150$

B1 for $12x + 8y = 1480$

M1 for a valid method to solve the equations (e.g., multiplying the first equation by 8 to get $8x + 8y = 1200$ and subtracting)

$x = 70$ and $y = 80$ **[A1]**

(b) M1 for isolating the square root: $\frac{T}{2\pi} = \sqrt{\frac{m}{k}}$

M1 for squaring both sides: $\frac{T^2}{4\pi^2} = \frac{m}{k}$

$\frac{kT^2}{4\pi^2}$ **[A1]**

[Total Marks: 7]

31 **(a)** $p(x+4) = 3x - 1$ **[M1]**

$px + 4p = 3x - 1$ **[M1]**

$3x - px = 4p + 1$ (or equivalent step isolating x terms on one side) **[M1]**

$\frac{4p+1}{3-p}$ (or $\frac{-4p-1}{p-3}$) **[A1]**

(b) $(2x \pm a)(x \pm b) = 0$ where $ab = -12$ or $2b \pm a = -5$ **[M1]**

$(2x+3)(x-4) = 0$ **[M1]**

$x = -1.5$, $x = 4$ **[A1]**

[Total Marks: 7]

32 **(a)** $2(3x + 2 + x - 1) = 42$ or $3x + 2 + x - 1 = 21$ **[M1]**

$4x + 1 = 21$ **[M1]**

5 **[A1]**

(b) $p^2 = \frac{x-3}{x}$ **[M1]**

$p^2 x = x - 3$ **[M1]**

$p^2 x - x = -3$ or $x - p^2 x = 3$ or $x(p^2 - 1) = -3$ **[M1]**

$\frac{3}{1-p^2}$ or $\frac{-3}{p^2-1}$ **[A1]**

[Total Marks: 7]

33 **(a)** Multiplies by denominator: $y(x-3) = 2x + 5$ **[M1]**

Expands and collects x terms on one side: $xy - 2x = 3y + 5$ **[M1]**

$\frac{3y+5}{y-2}$ or $\frac{-3y-5}{2-y}$ **[A1]**

(b) Multiplies throughout by common denominator $3(x-2)$ to reach $(x+1)(x-2) - 6 = 3(x-2)$ **[M1]**

Expands brackets correctly: $x^2 - 2x + x - 2 - 6 = 3x - 6$ **[M1]**

Rearranges to form a correct 3-term quadratic equation: $x^2 - 4x - 2 = 0$ **[A1]**

Uses the quadratic formula $\frac{-(-4) \pm \sqrt{(-4)^2 - 4(1)(-2)}}{2(1)}$ **[M1]**

4.45 **[A1]**

-0.45 **[A1]**

[Total Marks: 9]

34 **(a)** Sets up equation for area: $(2x+3)(x-1) = 45$ **[M1]**

Expands and simplifies to given result: $2x^2 - 2x + 3x - 3 = 45 \implies 2x^2 + x - 48 = 0$ **[A1]**

(b) Uses quadratic formula with correct values: $\frac{-1 \pm \sqrt{1^2 - 4(2)(-48)}}{2(2)}$ **[M1]**

Correct discriminant: $\frac{-1 \pm \sqrt{385}}{4}$ **[M1]**

4.655... and $-5.155...$ **[A1]**

4.66 and -5.16 **[A1]**

(c) Multiplies by denominator: $y(x-3) = ax+5$ **[M1]**

Expands brackets: $xy - 3y = ax + 5$ **[M1]**

Collects terms in x on one side: $xy - ax = 3y + 5$ **[M1]**

Factorises x and divides: $\frac{3y+5}{y-a}$ (or equivalent) **[A1]**

[Total Marks: 10]

35 $\frac{3(x-1)+1(x+2)}{(x+2)(x-1)} = 2$ or $3(x-1)+1(x+2) = 2(x+2)(x-1)$ **[M1]**

$3x - 3 + x + 2$ or $4x - 1$ **[M1]**

$2x^2 - 2x - 3 = 0$ or $-2x^2 + 2x + 3 = 0$ **[A2]**

$\frac{-(-2) \pm \sqrt{(-2)^2 - 4(2)(-3)}}{2(2)}$ or GDC equivalent to solve the 3-term quadratic **[M1]**

1.82 and -0.82 **[A1]**

[Total Marks: 6]

36 Multiplies by the common denominator to get $3(2x+1)+4(x-2) = 2(x-2)(2x+1)$ [M2]
(allow M1 for $\frac{3(2x+1)+4(x-2)}{(x-2)(2x+1)} = 2$)

Expands brackets to get $10x - 5 = 2(2x^2 - 3x - 2)$ or $10x - 5 = 4x^2 - 6x - 4$ **[M1]**

Rearranges to form a correct 3-term quadratic equation equal to zero: $4x^2 - 16x + 1 = 0$ **[A1]**

Substitutes correctly into the quadratic formula: $x = \frac{-(-16) \pm \sqrt{(-16)^2 - 4(4)(1)}}{2(4)}$ **[M1]**

$x = 3.94$ and $x = 0.06$ **[A1]**

[Total Marks: 6]

37 Multiplies by the common denominator to get $5(2x+1) - 3(x-2) = 2(x-2)(2x+1)$ **[M2]**

If not M2, M1 for a correct single fraction $\frac{5(2x+1)-3(x-2)}{(x-2)(2x+1)}$ seen

$4x^2 - 13x - 15 = 0$ **[A2]**

If not A2, A1 for correctly expanding sides to $7x + 11$ or $4x^2 - 6x - 4$

$x = \frac{-(-13) \pm \sqrt{(-13)^2 - 4(4)(-15)}}{2(4)}$ **[M1]**

4.15 **[A1]**

-0.90 **[A1]**

[Total Marks: 7]

38 **(a)** $\dfrac{3(x+3)-2(x-1)}{(x-1)(x+3)} = 1$ **[M1]**

$3x + 9 - 2x + 2 = (x-1)(x+3)$ **[M1]**

$x + 11 = x^2 + 2x - 3$ **[M1]**

$x^2 + x - 14 = 0$ established with no errors seen **[A1]**

(b) $\dfrac{-1 \pm \sqrt{1^2 - 4(1)(-14)}}{2(1)}$ **[M1]**

$\dfrac{-1 \pm \sqrt{57}}{2}$ **[M1]**

$3.27, -4.27$ **[A1]**

[Total Marks: 7]

39 **(a)** **(i)** Forms a correct initial equation involving time: $\dfrac{120}{x} - \dfrac{120}{x+10} = \dfrac{36}{60}$ **[M1]**

Simplifies the time difference to $\dfrac{3}{5}$ and multiplies by a common denominator: $600(x+10) - 600x = 3x(x+10)$ **[M1]**

$6000 = 3x^2 + 30x$ leading correctly to $x^2 + 10x - 2000 = 0$ with no errors **[A1]**

(ii) $(x+50)(x-40) = 0$ **[M1]**

$x = 40, x = -50$ **[A1]**

(b) Uses the quadratic formula with correct substitution: $x = \dfrac{-(-7) \pm \sqrt{(-7)^2 - 4(3)(-5)}}{2(3)}$ **[M1]**

Simplifies to $\dfrac{7 \pm \sqrt{109}}{6}$ **[M1]**

$x = 2.91$ or $x = -0.57$ **[A1]**

[Total Marks: 8]

40 **(a)** Constructs initial time equation: $\dfrac{120}{x} + \dfrac{150}{x+10} = 4$ **[M1]**

Multiplies by common denominator $x(x+10)$ to reach $120(x+10) + 150x = 4x(x+10)$ **[M1]**

Expands brackets correctly: $120x + 1200 + 150x = 4x^2 + 40x$ **[M1]**

Rearranges to $4x^2 - 230x - 1200 = 0$ and divides by 2 to yield $2x^2 - 115x - 600 = 0$ with no errors seen **[A1]**

(b) Substitutes values into the quadratic formula $\dfrac{115 \pm \sqrt{(-115)^2 - 4(2)(-600)}}{4}$ **[M1]**

62.3 **[A1]**

(c) -1.84 **[B1]**

2.36 **[B1]**

[Total Marks: 8]

41 (a) $\frac{40}{x}$ **[B1]**

(b) Converts 45 minutes to hours: $\frac{45}{60}$ or $\frac{3}{4}$ (or 0.75) **[B1]**

Forms a correct fractional equation: $\frac{40}{x} - \frac{40}{x+20} = \frac{3}{4}$ **[M1]**

Finds a common denominator or multiplies through to clear fractions:
$40(x+20) - 40x = \frac{3}{4}x(x+20)$ **[M1]**

Expands correctly and simplifies to the given quadratic with no errors: $800 = \frac{3}{4}(x^2 + 20x)$

$3200 = 3x^2 + 60x$

$3x^2 + 60x - 3200 = 0$ [A1]

(c) Correct substitution into the quadratic formula: $x = \frac{-60 \pm \sqrt{60^2 - 4(3)(-3200)}}{2(3)}$ **[M1]**

Correct simplification of the discriminant: $\frac{-60 \pm \sqrt{42000}}{6}$ **[M1]**

24.16 and -44.16 (both correct) **[A1]**

[Total Marks: 8]

42 (a) M1 for multiplying by the common denominator to obtain
$5(x-3) + 2(x+2) = (x+2)(x-3)$

M1 for expanding the brackets correctly: $5x - 15 + 2x + 4 = x^2 - x - 6$

M1 for rearranging into a correct three-term quadratic equation, e.g., $x^2 - 8x + 5 = 0$

M1 for correct use of the quadratic formula, e.g., $\frac{-(-8) \pm \sqrt{(-8)^2 - 4(1)(5)}}{2(1)}$

7.32 and 0.68 **[A1]**

(b) M1 for multiplying by the denominator: $y(5x+1) = 2x - 3$

M1 for expanding the bracket and collecting x terms on one side: $5xy - 2x = -y - 3$ or $2x - 5xy = y + 3$

M1 for factorising x: $x(5y-2) = -y - 3$ or $x(2-5y) = y + 3$

$\frac{-y-3}{5y-2}$ or $\frac{y+3}{2-5y}$ **[A1]**

[Total Marks: 9]

43 (a) $x(x-6) = 3(x+2)$ **[M1]**

$x^2 - 6x = 3x + 6$ **[M1]**

$x^2 - 9x - 6 = 0$ **[A1]**

Uses the quadratic formula, e.g., $x = \dfrac{-(-9) \pm \sqrt{(-9)^2 - 4(1)(-6)}}{2(1)}$ **[M1]**

$\dfrac{9 \pm \sqrt{105}}{2}$ **[A1]**

(b) $y(3x + 1) = 2x - 5$ **[M1]**

$3xy + y = 2x - 5$ **[M1]**

Isolates terms containing x on one side, e.g., $3xy - 2x = -y - 5$ or $2x - 3xy = y + 5$ **[M1]**

$\dfrac{-y - 5}{3y - 2}$ or $\dfrac{y + 5}{2 - 3y}$ **[A1]**

[Total Marks: 9]

44 (a) $3(2x + 1) + 2(x - 1) = 2(x - 1)(2x + 1)$ **[M1]**

$6x + 3 + 2x - 2 = 4x^2 - 2x - 2$ **[M1]**

$4x^2 - 10x - 3 = 0$ **[A1]**

$\dfrac{10 \pm \sqrt{(-10)^2 - 4(4)(-3)}}{2(4)}$ **[M1]**

$2.77, -0.27$ **[A1]**

(b) $x^2 + (x + 2)^2 = 340$ (or equivalent setup for consecutive even numbers) **[M1]**

$2x^2 + 4x - 336 = 0$ or $x^2 + 2x - 168 = 0$ **[A1]**

$(x + 14)(x - 12) = 0$ or correct application of quadratic formula **[M1]**

12 and 14 (both correct) **[A1]**

[Total Marks: 9]

45 (a) $\dfrac{100}{x} - \dfrac{100}{x + 15} = \dfrac{1}{2}$ (or equivalent e.g., $\dfrac{100}{x} = \dfrac{100}{x + 15} + 0.5$) **[M1]**

$200(x + 15) - 200x = x(x + 15)$ **[M1]**

$200x + 3000 - 200x = x^2 + 15x$ leading to $x^2 + 15x - 3000 = 0$ **[A1]**

(b) $\dfrac{-15 \pm \sqrt{15^2 - 4(1)(-3000)}}{2(1)}$ **[M2]**

Allow [M1] for correctly substituting values into the quadratic formula or evaluating the discriminant $\sqrt{12225}$

47.78 **[A1]**

-62.78 **[A1]**

(c) Speed of car $= 47.78 + 15 = 62.78$ km/h **[M1]**

Time $= \dfrac{100}{62.78} = 1.592...$ hours **[M1]**

1 hour 36 minutes **[A1]**

[Total Marks: 10]

46 **(a)** Multiplies by the denominator to obtain $y(x+4) = 3x - 2$ **[M1]**

Expands the bracket to obtain $xy + 4y = 3x - 2$ **[M1]**

Collects terms in x on one side, e.g., $xy - 3x = -4y - 2$ or $3x - xy = 4y + 2$ **[M1]**

Factorises x out and divides: $\dfrac{-4y-2}{y-3}$ or $\dfrac{4y+2}{3-y}$ **[A1]**

(b) Writes with common denominator or multiplies through to eliminate fractions:
$5(2x+1) - 3(x-2) = 2(x-2)(2x+1)$ **[M1]**

Correctly expands LHS: $10x + 5 - 3x + 6$ **[M1]**

Correctly expands RHS: $2(2x^2 - 3x - 2)$ or $4x^2 - 6x - 4$ **[M1]**

Rearranges to form a correct 3-term quadratic equation, e.g., $4x^2 - 13x - 15 = 0$ **[A1]**

Correctly substitutes into the quadratic formula: $\dfrac{-(-13) \pm \sqrt{(-13)^2 - 4(4)(-15)}}{2(4)}$ **[M1]**

$x = 4.15$ and $x = -0.90$ **[A1]**

[Total Marks: 10]

47 **(a)** $y(x+2) = 3x - 1$ **[M1]**

$yx + 2y = 3x - 1$ **[M1]**

$2y + 1 = 3x - yx$ or $yx - 3x = -2y - 1$ **[M1]**

$x = \dfrac{2y+1}{3-y}$ or $x = \dfrac{-2y-1}{y-3}$ **[A1]**

(b) Multiply throughout by $(x-1)(x+2)$: $2(x+2) + 3(x-1) = (x-1)(x+2)$ **[M1]**

Expand terms: $2x + 4 + 3x - 3 = x^2 + x - 2$ **[M1]**

Collect to form a quadratic equation: $x^2 - 4x - 3 = 0$ **[A1]**

Use the quadratic formula: $x = \dfrac{-(-4) \pm \sqrt{(-4)^2 - 4(1)(-3)}}{2(1)}$ **[M1]**

$x = \dfrac{4 \pm \sqrt{28}}{2}$ **[M1]**

$x = 4.65$ and $x = -0.65$ **[A1]**

[Total Marks: 10]

48 **(a)** $\dfrac{150}{x}$ **[B1]**

(b) $\dfrac{150}{x-15} - \dfrac{150}{x} = 0.5$ (or $\dfrac{30}{60}$ or $\dfrac{1}{2}$) **[M1]**

$150x - 150(x-15) = 0.5x(x-15)$ **[M1]**

$150x - 150x + 2250 = 0.5x^2 - 7.5x$ **[M1]**

$4500 = x^2 - 15x \implies x^2 - 15x - 4500 = 0$ with no errors seen **[A1]**

(c) $(x-75)(x+60) = 0$ or $\dfrac{-(-15) \pm \sqrt{(-15)^2 - 4(1)(-4500)}}{2(1)}$ **[M2]**

(or $M1$ for correct factors $(x+a)(x+b)$ where $ab = -4500$ or $a + b = -15$, or for correct

substitution into quadratic formula)

75 (rejects negative solution -60) **[A1]**

(d) $P(q-1) = 2q+3$ **[M1]**

$Pq - P = 2q + 3$ **[M1]**

$Pq - 2q = P + 3$ **[M1]**

$\frac{P+3}{P-2}$ or $\frac{-P-3}{2-P}$ **[A1]**

[Total Marks: 12]